

Ziseok Lee



Personal Website: <https://ziseoklee.github.io/>
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Integrated Masters with PhD Program
Department of Biomedical Sciences
Seoul National University

Research Interests

Generative Models, AI4Science, Computer Vision

Education

Seoul National University	2025.03 - present
Masters-PhD Program in Biomedical Sciences advised by Prof. Kyungsu Kim	
Seoul National University	2021.03 - 2025.02
B.S. in Mathematical Sciences (Major), B.S. in Computer Science and Engineering (Double Major)	
GPA 4.22/4.30 (Summa Cum Laude)	

Research Experience

▼ Student Researcher at AIBL	2024.10 - present
Probabilistic Generative Models for Biomolecule Generation and Drug Design	
▼ Undergraduate Research Assistant at SNU MLLAB	2024.07 - 2024.09
Blackbox Optimization	
▼ Undergraduate Research Assistant at CTA Lab	2024.01 - 2024.08
NP-hard Subgraph Matching Algorithms with Safety Conditions for Reduced Candidate Space	
▼ Undergraduate Research Assistant at HYKE Group	2022.12 - 2023.07
Collective Motion in the Vicsek Particle Model, Hyperbolic Conservation Laws with Nonlocal Relaxation	

Awards & Scholarships

AI Seoul Tech Graduate Scholarship (Seoul Scholarship Foundation)	2025
B.S. Degree Honors: Summa Cum Laude (Seoul National University)	2025
Presidential Science Scholarship (Korea Student Aid Foundation)	2023-2024
Dean's List (Department of Mathematical Sciences, SNU)	2023-2024
Professor Heo Sik Scholarship (Department of Mathematical Sciences, SNU)	2022
Gwanak Foundation Scholarship (Gwanak Foundation)	2021

Teaching Experience

▼ Teaching Assistant for [Data Mining and Machine Learning]	2026.03 - 2026.06
Assisted Professor Kyungsu Kim in teaching a third-year undergraduate course, supporting students through discussion sessions.	
▼ Teaching Assistant for [Introduction to Data Science]	2025.03 - 2025.06

Assisted Professor Kyungsu Kim in teaching a second-year undergraduate course, supporting students through discussion sessions.	
▼ Teaching Assistant for [Calculus 1]	2024.03 - 2024.06
Taught a course on Calculus 1. Was awarded Excellent TA and gave a case presentation.	
▼ Student-Directed Seminar [Understanding the Brain as a Complex System]	2023.09 - 2023.12
Organized and taught a student-directed seminar integrating theoretical neuroscience, network science, deep learning, biology, and psychology to understand the brain as a complex, dynamical, entangled system.	
▼ Peer Tutoring [Calculus Tutoring for First Year Students]	2023.03 - 2023.06
Taught a tutoring class for "Differential and Integral Calculus 1" during the Spring semester of 2023.	

Grants & Funding

▼ Strategic Hub for International Research Collaboration (SNU Office of Research Affairs)	2026 - 2028
Project Title: An International Research Hub for Developing a 4D Molecular Generative AI Platform to Control Dynamic Protein Interaction Networks.	
Funding: ~\$380k (3 Years)	
Description: Authored the grant proposal and secured funding for a 3-year international joint research project between SNU and Harvard Medical School (Prof. Kwang-Soo Kim). I lead the design of stochastic interpolant-based diffusion models for drug optimization and protein conformational ensemble prediction.	

Publications

* Equal contribution, † Corresponding Author

Conference Papers

- [First Author]** *On the Collapse of Generative Paths: A Criterion and Correction for Diffusion Steering*, Ziseok Lee*†, Minyeong Hwang*, Wooyeol Lee, Sanghyun Jo, Jihyung Ko, Young Bin Park, Jae-Mun Choi, Eunho Yang†, Kyungsu Kim†, Forty-third International Conference on Machine Learning (ICML) (2026)
- [Co-first Author]** *TRACE: Your Diffusion Model is Secretly an Instance Edge Detector*, Sanghyun Jo*, Ziseok Lee*, Wooyeol Lee, Jonghyun Choi, Jaesik Park†, Kyungsu Kim†, The Fourteenth International Conference on Learning Representations (**ICLR Oral, top 1.1%**) (2026)
- [Co-first Author]** *Early Timestep Zero-Shot Candidate Selection for Instruction-Guided Image Editing*, Joowon Kim*, Ziseok Lee*, Donghyeon Cho, Sanghyun Jo, Yeonsung Jung, Kyungsu Kim†, Eunho Yang†, Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (2025)

Workshop Papers

- [Coauthor]** *HybridLinker: Topology-Guided Posterior Sampling for Enhanced Diversity and Validity in 3D Molecular Linker Generation*, Minyeong Hwang, Ziseok Lee, Kwang-Soo Kim, Kyungsu Kim†, Eunho Yang†, ICML 2025 Generative AI and Biology (GenBio) Workshop (2025)

Preprints

- [Co-first Author]** *Value-Informed Schedule Trimming for Accelerated Sequential Monte Carlo Guidance in Discrete Diffusion*, Jaehyeon Kim*, Ziseok Lee*, Siyoung Choi, Kyungsu Kim†, (2026)
- [Co-first Author]** *ISAC: Training-Free Instance-to-Semantic Attention Control for Improving Multi-Instance Generation*, Sanghyun Jo*, Wooyeol Lee*, Ziseok Lee*, Kyungsu Kim†, arXiv preprint arXiv:2505.20935 (2025)

Academic Services

[2025.04] Assisted Professor Kyungsu Kim in reviewing submissions in the field of computer vision for ICCV 2025.

[2025.03] Assisted Professor Kyungsu Kim in reviewing two submissions in the field of flow matching for ICML 2025.

Biography

Ziseok Lee is a graduate student in the [Department of Biomedical Sciences](#) at [Seoul National University](#). He is a member of the [Artificial Intelligence and Biomedical Informatics Lab \(AIBL\)](#) under the supervision of Professor Kyungsu Kim. He received his B.S. degree in the [Department of Mathematical Sciences](#) with a double major in the [Department of Computer Science and Engineering](#) at Seoul National University in 2025. His research interests include generative models (e.g., stochastic interpolants) and their applications to computer vision and biomedical science, particularly molecule generation for drug design.

Technical Skills

Programming Languages: Python, C/C++, Java, JavaScript, HTML/CSS

Frameworks & Libraries: PyTorch, Scikit-learn, OpenCV, React

Tools & Platforms: Git, Docker, Linux

Languages: Korean (native), English (fluent)

References

Dr. Kyungsu Kim (Primary advisor)

Professor, School of Transdisciplinary Innovations

Seoul National University

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